

Grade 6
Unit 11
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

From the specified set of integers, determine which, if any, make the equation true.

$\{-10, 1, 3, 12\}$

1. $20 - 5w = 10w + 5$
 $20 - 5(-10) = 10(-10) + 5$
 $20 + 50 = -100 + 5$
 $70 \neq -95$ NOT TRUE

$20 - 5(1) = 10(1) + 5$
 $20 - 5 = 10 + 5$
 $15 = 15$ TRUE

$20 - 5(3) = 10(3) + 5$
 $20 - 15 = 30 + 5$
 $5 \neq 35$ NOT TRUE

$20 - 5(12) = 10(12) + 5$
 $20 - 60 = 120 + 5$
 $-40 \neq 125$ NOT TRUE

3. $3y + 5 = 4y + 15$
 $3(-10) + 5 = 4(-10) + 15$
 $-30 + 5 = -40 + 15$
 $-25 = -25$ TRUE

$3(1) + 5 = 4(1) + 15$
 $3 + 5 = 4 + 15$
 $8 \neq 19$ NOT TRUE

$3(3) + 5 = 4(3) + 15$
 $9 + 5 = 12 + 15$
 $14 \neq 27$ NOT TRUE

$3(12) + 5 = 4(12) + 15$
 $36 + 5 = 48 + 15$
 $41 \neq 63$ NOT TRUE

2. $4 + x = 2x - 8$
 $4 + (-10) = 2(-10) - 8$
 $-6 = -20 - 8$
 $-6 \neq -28$ NOT TRUE

$4 + (1) = 2(1) - 8$
 $5 = 2 - 8$
 $5 \neq -6$ NOT TRUE

$4 + (3) = 2(3) - 8$
 $7 = 6 - 8$
 $7 \neq -2$ NOT TRUE

$4 + (12) = 2(12) - 8$
 $16 = 24 - 8$
 $16 = 16$ TRUE

4. $2p - 6 = 6p - 18$
 $2(-10) - 6 = 6(-10) - 18$
 $-20 - 6 = -60 - 18$
 $-26 \neq -78$ NOT TRUE

$2(1) - 6 = 6(1) - 18$
 $2 - 6 = 6 - 18$
 $-4 \neq -12$ NOT TRUE

$2(3) - 6 = 6(3) - 18$
 $6 - 6 = 18 - 18$
 $0 = 0$ TRUE

$2(12) - 6 = 6(12) - 18$
 $24 - 6 = 72 - 18$
 $18 \neq 54$ NOT TRUE

Determine if the inequality $18 < 2m - 6$ is true for each value.

5. 8
 $18 < 2(8) - 6$
 $18 < 16 - 6$
 $18 < 10$
NOT TRUE

6. 12
 $18 < 2(12) - 6$
 $18 < 24 - 6$
 $18 < 18$
NOT TRUE

7. 15
 $18 < 2(15) - 6$
 $18 < 30 - 6$
 $18 < 24$
TRUE

Determine if the inequality $-y + 3 + 8y > -3y - 7$ is true for each value.

8. -5
 $-(-5) + 3 + 8(-5) > -3(-5) - 7$
 $5 + 3 - 40 > 15 - 7$
 $-32 > 8$
NOT TRUE

9. 0
 $-(0) + 3 + 8(0) > -3(0) - 7$
 $0 + 3 + 0 > 0 - 7$
 $3 > -7$
TRUE

10. 5
 $-(5) + 3 + 8(5) > -3(5) - 7$
 $-5 + 3 + 40 > -15 - 7$
 $38 > -22$
TRUE